higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

31 March 2026

This marking guideline consists of 8 pages.

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TVET Exam Papers NATED & NCV	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	**N4** Mechanical Draughting NATED
N1-N3 Electrical Trade Theory NATED	**N2-N3** Diesel Trade Theory NATED	**N5-N6** Strength of Materials & Structures NATED	**N3** Building & Civil Technology NATED	**N1-N2** Fitting and Machining Theory NATED	**N1-N4** Plating and Structural Steel Drawing NATED
N1-N3 Motor Trade Theory NATED	**N5-N6** Mechanical Drawing & Design NATED	**N1-N2** Plumbing Theory NATED	**N4-N6** Fault Finding & Protective Devices NATED	**N2-N3** Instrument Trade Theory NATED	**N5-N6** Engineering Physics NATED
N5-N6 Fluid Mechanics NATED	**N4-N6** Supervisory Management NATED	**N4-N6** Digital Electronics NATED	**N6** Control Systems NATED	**N1-N2** Bricklaying &Plastering Theory NATED	More…

QUESTION 1: AC THEORY

1.1 1.1.1 Testing
1.1.2 Harmonics
1.1.3 Observed
1.1.4 Faulty circuit
1.1.5 Existence

(5 × 1) (5)

1.2 1.2.1

$$Z_D = \frac{L}{RC}$$

$$C = \frac{L}{R Z_D} \checkmark$$

$$C = \frac{110 \ x \ 10^{-6}}{20 \times 21713}$$

$$C = 253{,}3 \ pF \checkmark$$ (3)

1.2.2 10 kΩ (1)

1.3 1.3.1

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

$$Q = \frac{1}{29}\sqrt{\frac{0{,}4}{56 \ x \ 10^{-6}}} \checkmark$$

$$Q = 2{,}914$$ (2)

1.3.2

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$= \frac{1}{2\pi \times \sqrt{0{,}4 \ \times \ 56 \ x \ 10^{-6}}} \checkmark$$

$$= 33{,}628 \ Hz$$

$$BW = \frac{f_r}{Q}$$

$$BW = \frac{33{,}628}{2{,}914}$$

$$BW = 11{,}54 \ Hz$$ (4)

[15]

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QUESTION 2: POWER SUPPLIES

2.1 2.1.1 The change in output voltage for a line supply voltage,✓ keeping the load current and temperature constant✓

2.1.2 The variation of the DC output voltage✓ as a function of the DC load current✓

(2 × 2) (4)

2.2 $V_{NL} = V_{DC} = I_{DC} R_{DC}$

$V_{DC} = 100 \ x \ 10^{-3} \times 170$✓

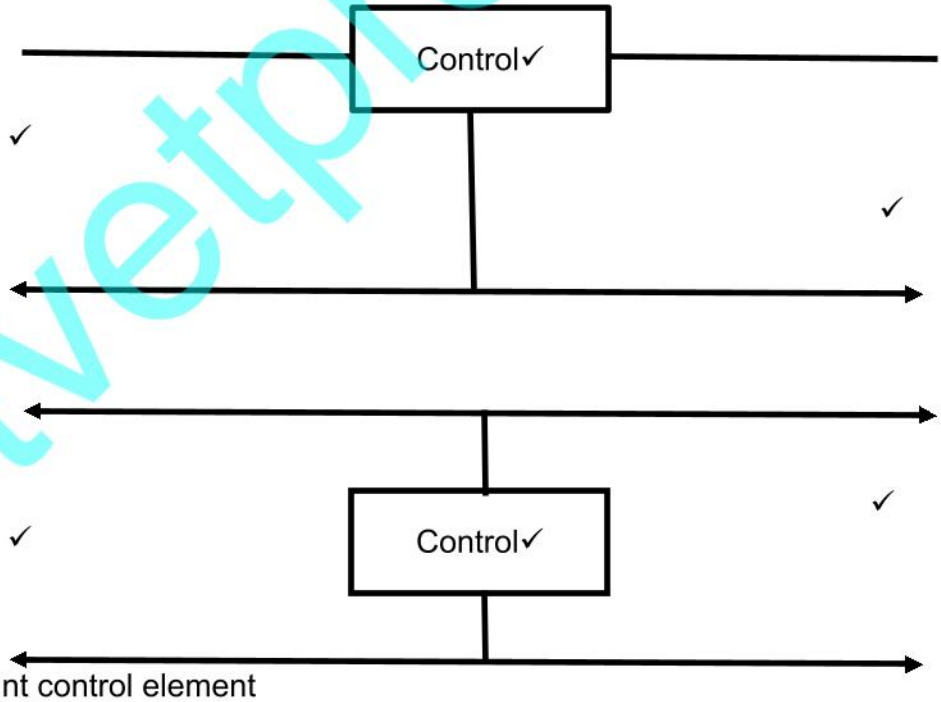
$= 17 \ V$

$\% \ VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$

$= \frac{17 - 13}{13} \times 100$

$= 30{,}77\%$✓ (4)

2.3 Series control element



(6)

2.4 To increase the current supply rating to the load (1)

[15]

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QUESTION 3: AMPLIFIERS

3.1 To limit the collector current during an increase in temperature (1)

3.2 3.2.1 $V_E = I_C R_E$

$= 10\ x\ 10^{-3} \times 130$✓

$= 1{,}3\ V$✓

$V_{CC} = 10 V_E$

$V_{CC} = 10 \times 1{,}3$✓

$V_{CC} = 13\ V$✓ (4)

3.2.2 $V_B = V_{BE} + V_E$

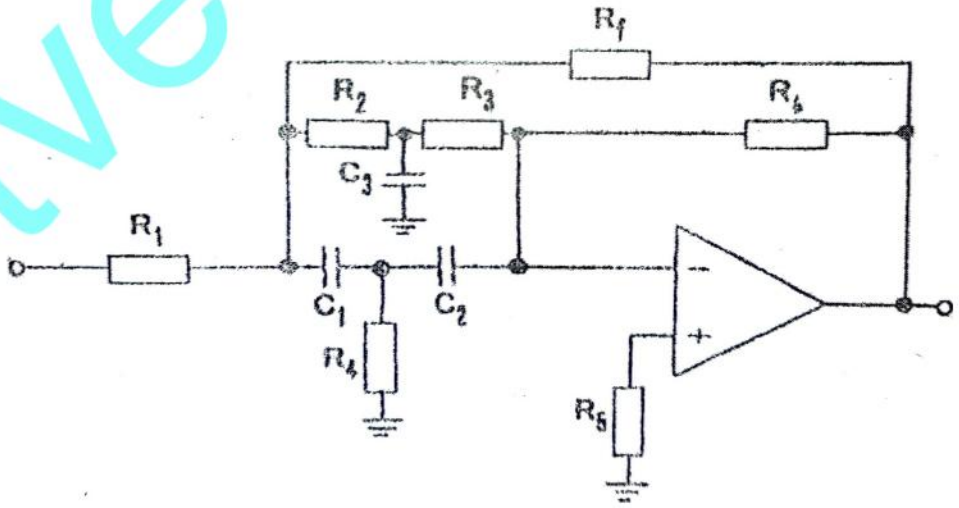
$V_{BE} = V_B - V_E$

$= 1{,}9 - 1{,}3$✓

$= 0{,}6\ V$✓ (2)

3.3
- Temperature variations
- Unstable voltage supply
- Parameter variations of components
- Aging of components
- Changing load conditions (Any 4 × 1) (4)

3.4



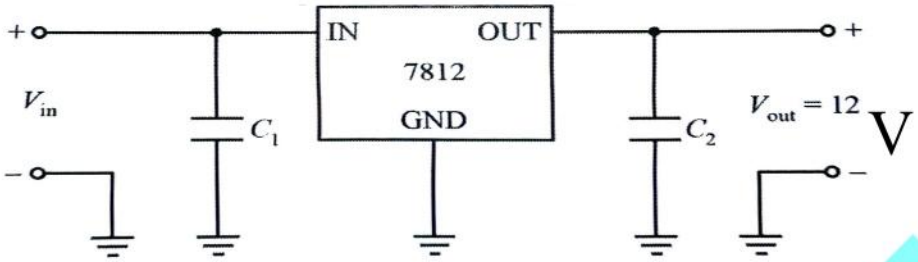
(ONE full mark for operational amplifier and half marks for any SIX correctly labelled electric components) (4)

[15]

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QUESTION 4: INTERGRATED CIRCUITS AND TRANSDUCERS

4.1



(ONE full mark for regulator and half marks for any FOUR correctly labelled electric components) (3)

4.2
- Connect all test equipment to earth.
- Switch off the power before you work on the circuit.
- Use a special earth wristband. (3)

4.3

$1500 = A e^{\frac{B}{284}} \ldots\ldots\ldots\ldots\ldots\ldots (1)$

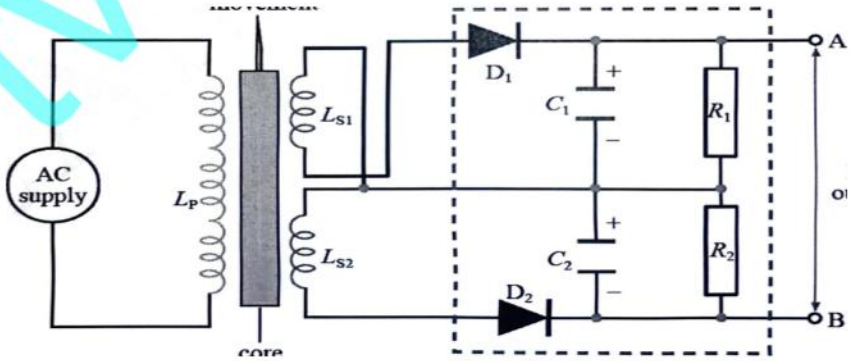
$15\,000 = A e^{\frac{B}{308}} \ldots\ldots\ldots\ldots\ldots\ldots (2)$ ✓

$(2) \div (1)$ $\qquad 10 = B\left(\frac{1}{284} - \frac{1}{308}\right) \times e$ ✓

$$B = \frac{\ln 10}{\left(\frac{1}{284} - \frac{1}{308}\right)}$$ ✓

$= 8\,392{,}155$ ✓ (4)

4.4



co

(ONE full mark for transformer and half marks for any SIX correctly labelled electric components) (3)

re

4.5
- Sensitive to magnetic fields (2)

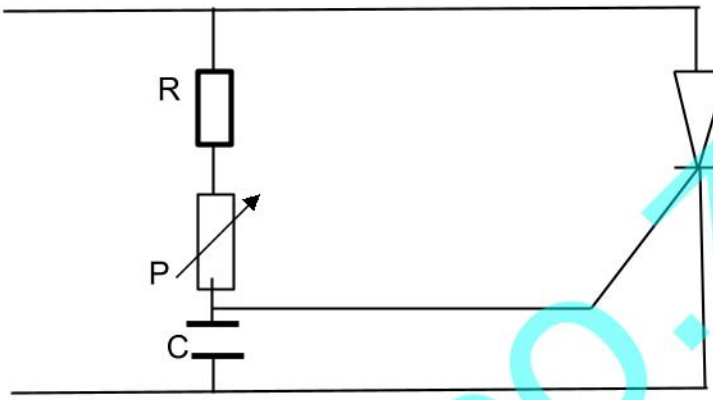
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- Operating range will be reduced due to side effects (Any relevant 2 × 1)

[15]

QUESTION 5

5.1 5.1.1

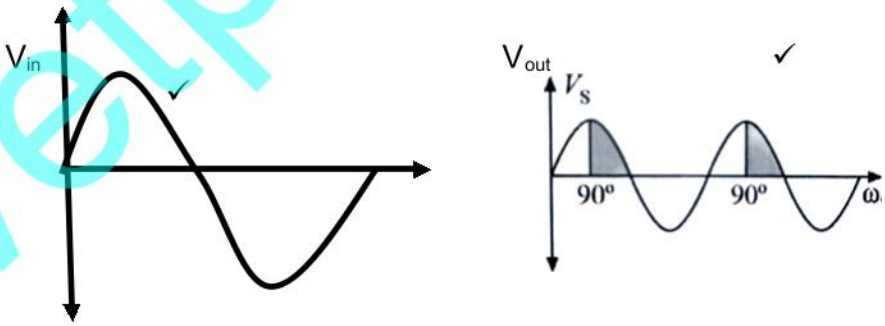


(ONE half mark for every correctly labelled electric component)

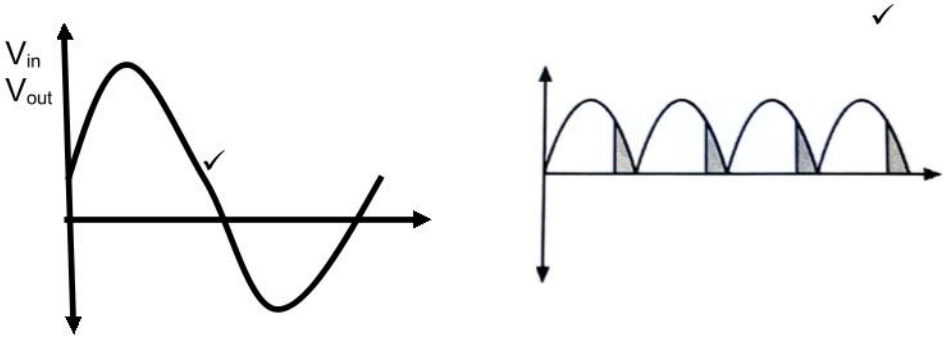
(2)

5.1.2 As soon as the capacitor voltage reaches the required gate voltage, the thyristor switches on.✓ The thyristor will stay switched on until the bias drops to such an extent that the current is less than the holding current value.✓ This period is called the conductor angle.✓ If the potentiometer is used, the phase angle can be adjusted to the required value.✓ (4)

5.2 5.2.1



5.2.2

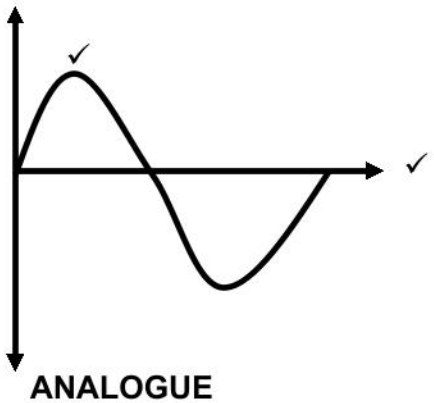


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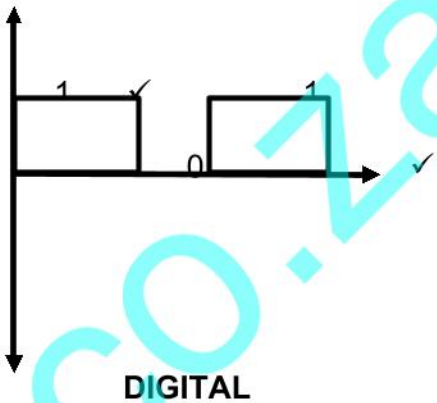
(2 × 2) (4)
[10]

QUESTION 6: MEASURING INSTRUMENTS

6.1



ANALOGUE



DIGITAL (4)

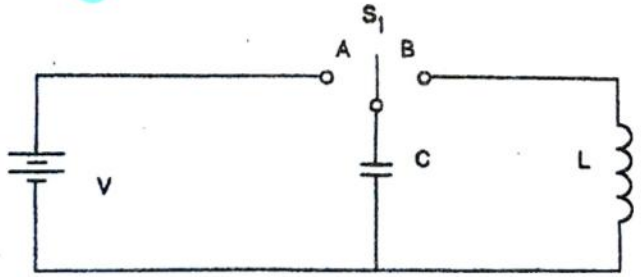
6.2
- The input voltage generates a charging current, which charges the capacitor until it reaches the reference voltage.
- Once Vr is attained, the comparator changes its state, activating the precision pulse generator.
- The pulse generator generates pulse with precise charge content, swiftly discharging the capacitor.
- The rate at which the capacitor charges and discharges results in a signal frequency directly proportional to the input voltage. (Any 3 × 1) (3)

6.3
- Direct compensation
- Voltage-to-time conversion
- Voltage to frequency (3)

[10]

QUESTION 7: OSCILLATORS

7.1



(ONE half mark for any FOUR correctly labelled electric components)

If the switch is close to A, the capacitor charges to Vm.✓ If the switch is close (5)

to B, the capacitor will discharge through the inductor and the discharged current produces a magnetic field around the inductor while the capacitor is discharging.✓ After the capacitor discharge, the current flow ceases and the magnetic line collapses, inducing a back EMF in the inductor in the opposite direction to the voltage applied by the capacitor. This process repeats itself until the oscillation lapses.✓

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7.2
- Hartley
- Colpitts
- RC phase shift
- Crystal controller
- Wien bridge

(Any 3 × 1) (3)

7.3 An electronic connector✓ that uses positive feedback✓ (2)
[10]

QUESTION 8: LIQUID CRYSTAL DISPLAY

8.1 When an electrical current is applied to the liquid crystal molecule, the molecule tends to untwist.✓ This causes the angle of light that passes through the molecule of the polarised glass✓ and causes a charge in the angle of the polarising fitter.✓ As a result, a little light is allowed to pass✓ the polarised glass through a particular area of the LCD.✓ (5)

8.2
- Low reliability
- Speed is very low
- Requires an AC drive
- Requires additional light source
- Range of temperature is limited for operation

(5)
[10]

TOTAL: 100

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